The Cone Generated by Positive Semidefinite Mean Polynomials

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Abstract

Certain classes of homogeneous polynomials, such as Positive Semidefinite (PSD) Diagonal-minus-Tail (D-T) forms and those derived from the Arithmetic-Geometric Mean inequality—specifically Sums of Nonnegative Circuit (SONC) polynomials—can be identified as nonnegative mean polynomials. This paper studies PSD forms defined as weighted power means, denoted by $M_{q,p}(Y, w)$, and investigates the cone of nonnegative mean real n-variate forms of degree $2d$, denoted by $\mathcal{M}_{n,2d}(X, \alpha)$. We analyze the relationship between $\mathcal{M}_{n,2d}$ and other fundamental cones in real algebraic geometry: the cone of all PSD forms ($\mathcal{P}_{n,2d}$), the cone of Sums of Squares ($\Sigma_{n,2d}$), and the SONC cone ($\mathcal{C}_{n,2d}$). Furthermore, we examine the structural properties of the sum of the SOS and SONC cones, commonly referred to as the SOSONC cone.

1 Introduction

The certification of nonnegativity for multivariate polynomials is a central problem in real algebraic geometry ...with significant applications in polynomial optimization. While deciding whether a general polynomial is Positive Semidefinite (PSD) is NP-hard, specific sub-classes of polynomials offer tractable certificates of nonnegativity. Notable examples include Sums of Squares (SOS), Positive Semidefinite Diagonal-minus-Tail (PSD D-T) forms, and Sums of Nonnegative Circuit (SONC) polynomials, which are derived from the Arithmetic-Geometric Mean inequality.

In this paper, we propose a unified framework for these classes by identifying them with *nonnegative mean polynomials*. We study PSD forms that are constructed as weighted power means, denoted by $M_{q,p}(Y, w)$, where $Y = (Y_1, ..., Y_m)$ is a tuple of symbols and $w \in \mathbb{R}^m_+$ is a vector of positive weights.

Specifically, we investigate the **cone of nonnegative mean real n-variate forms of degree** $2d$, denoted as $\mathcal{M}_{n,2d}(X, \alpha)$, where $X = (X_1, ..., X_n)$ and $\alpha \in \mathbb{N}^n_0$. A primary objective of this work is to elucidate the relationship between $\mathcal{M}_{n,2d}$ and the following established cones:

- $\mathcal{P}_{n,2d}$: The cone of all positive semidefinite (PSD) forms. Note that by definition, $\mathcal{M}_{n,2d} \subseteq \mathcal{P}_{n,2d}$.
- $\Sigma_{n,2d}$: The cone of sums of squares (SOS) polynomials.
- $\mathcal{C}_{n,2d}$: The cone of sums of nonnegative circuit (SONC) polynomials.
- **SOSONC**: The sum of the SOS and SONC cones, defined by M. Schick as $(\Sigma + \mathcal{C})_{n,2d}$.

Structure of the Paper

The remainder of this paper is organized as follows:

In §2, we provide the preliminaries on mean polynomials. We introduce the weighted power means $M_{q,p}(Y, w)$ for $q, p \in \mathbb{N}_0$, and utilizing Jensen's inequality, we prove that if $p < q$, the form $M_{q,p}$ is PSD (Proposition 1). We further demonstrate how standard classes of nonnegative polynomials can be recovered as instances of these means.

In §3, we revisit nonnegative circuit polynomials. We provide necessary and sufficient conditions for the nonnegativity of D-T forms and circuit polynomials using the language of mean polynomials, recovering established results found in the literature. We formally define the SONC cone $\mathcal{C}_{n,2d}$ (Definition 1) and show that the circuit nonnegativity criterion (Theorem 2) implies the D-T criterion (Theorem 1) as a special case; the two results are not equivalent.

In §4, we investigate the relationship between mean polynomials and Sums of Squares. We study $M_{2d,p}(X, \alpha)$ with $|\alpha| = 2d$ and $p \mid 2d$. We prove that $M_{2d,0}$ forms are Sums of Binomial Squares (SOBS) (Proposition 2). Additionally, for cases $d = 1, 2, 3$, we prove via induction and reduction to Hilbert's theorem that these forms are SOS for all $n \in \mathbb{N}$ (Theorem 3).

In §5, we move beyond the SOSONC cone to understand the general form $M_{q,p}(Y, w)$. We examine the Choi-Lam form Q and the second Robinson form $\hat{R}$. While Q is known to be SONC but not SOS, we establish a representation of Q and $\hat{R}$ as mean polynomials—specifically showing that $\hat{R}$ can be expressed as $8M_{1,0}(G, \mathbf{1})$ for a specific linear transformation G. This identifies $\hat{R}$ as a separating form for the cone of mean polynomials and the SOSONC cone.

The topics of Positivestellensatz, polynomial optimization, and differential-algebra/KKT objectives (§6–§7) are reserved for a follow-up paper.

2 Preliminaries on Mean Polynomials

In this section, we introduce the concept of weighted power means and establishing the fundamental inequalities that allow us to construct nonnegative polynomials.

Let $Y = (Y_1, \ldots, Y_m)$ be an m-tuple of variables (or symbols) and $w = (w_1, \ldots, w_m) \in \mathbb{R}_+^m$ be a vector of positive weights. We denote the power mean of order p as $M_p(Y, w)$.

2.1 Jensen's Inequality and Monotonicity

The nonnegativity of the mean polynomials studied in this paper relies on the monotonicity of the power mean function. We prove this using Jensen's inequality.

Proposition 1. *If $p < q$, then $M_p(Y, w) \leq M_q(Y, w)$ for any non-negative input Y.*

Proof. Consider the function $\varphi(x) = x^{\frac{2q}{p}}$ with $q > p$. The second derivative of this function is:

$$\varphi''(x) = \frac{2q}{p}\left(\frac{2q}{p} - 1\right) x^{\frac{2q}{p}-2}$$

Since $q > p$, the coefficient $\frac{2q}{p}(\frac{2q}{p} - 1)$ is strictly positive. Consequently, $\varphi(x)$ is a convex function on $\mathbb{R}_+$.

By Jensen's inequality, for any convex function φ and weights w_i, we have:

$$\varphi\left(\frac{\sum w_i x_i}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i)}{\sum w_i}$$

Substituting x_i with x_i^p in the inequality above, we obtain:

$$\varphi\left(\frac{\sum w_i x_i^p}{\sum w_i}\right) \leq \frac{\sum w_i \varphi(x_i^p)}{\sum w_i}$$

Substituting the definition $\varphi(t) = t^{q/p}$ (noting that the factor of 2 in the degree is preserved in the structure of the resulting forms), this inequality simplifies to:

$$\left(\frac{\sum w_i x_i^p}{\sum w_i}\right)^{\frac{q}{p}} \leq \frac{\sum w_i x_i^q}{\sum w_i}$$

Taking the q-th root, this is equivalent to $M_p(Y, w) \leq M_q(Y, w)$. □

2.2 The Cone of Mean Polynomials

Using the inequality established above, we can define a class of nonnegative polynomials. By raising both sides of the inequality $M_p \leq M_q$ to the power $c = \mathrm{lcm}(q, p)$, we eliminate rational exponents. We define the **mean polynomial form** $\mathcal{M}_{q,p}$ as:

$$\mathcal{M}_{q,p}(Y, w) = M_q^c(Y, w) - M_p^c(Y, w)$$

By Proposition 1, $\mathcal{M}_{q,p}$ is a nonnegative polynomial function in variables Y for all $q > p$.

Properties of Power Means

We recall the following basic limiting properties of the mean function $M_p(x, w)$ that are relevant to the boundaries of these cones:

- **Minimum:** $\lim_{p\to-\infty} M_p(x,w) = \min(x_1,\dots,x_n)$.
- **Maximum:** $\lim_{p\to\infty} M_p(x,w) = \max(x_1,\dots,x_n)$.
- **Geometric Mean:** $\lim_{p\to 0} M_p(x,w) = (\prod_{i=1}^n x_i^{w_i})^{1/\sum w_i}$.

Finally, we note that the equality $M_p(x,w) = M_q(x,w)$ holds if and only if all elements are equal, i.e., $x_1 = x_2 = \cdots = x_n$.

3 Nonnegative Diagonal minus Tail and Circuit Polynomials

In this section, we investigate the conditions for nonnegativity of Diagonal-minus-Tail (D-T) forms and general Circuit polynomials. We utilize the language of mean polynomials to re-derive established results found in [5] and [1].

Theorem 1. *Let $f(x)$ be a Diagonal-minus-Tail form defined as:*

$$f(x) = \sum_{i=1}^n \beta_i x_i^{2d} - \mu x^\alpha$$

where:

- $x = (x_1,\dots,x_n) \in \mathbb{R}^n$*;*
- *coefficients satisfy* $\beta_i, \mu > 0$*;*
- *exponents satisfy* $\alpha = (\alpha_1,\dots,\alpha_n) \in \mathbb{N}_0^n$ *and*

$$\sum_{i=1}^n \alpha_i = 2d.$$

The function $f(x)$ is positive semidefinite (PSD) on $\mathbb{R}_+^n$ if and only if:

$$\mu^{2d} \prod_{i=1}^n \alpha_i^{\alpha_i} \le (2d)^{2d} \prod_{i=1}^n \beta_i^{\alpha_i}$$

Proof. The proof relies on the weighted Arithmetic Mean-Geometric Mean (AM-GM) inequality.
($\Leftarrow$) Assume the condition holds. To show $f(x) \ge 0$, it suffices to show $\sum_{i=1}^n \beta_i x_i^{2d} \ge \mu x^\alpha$. Apply the weighted AM-GM inequality to the terms $y_i = \frac{\beta_i}{\alpha_i} x_i^{2d}$ with weights $w_i = \alpha_i$ (noting $\sum w_i = 2d$). We obtain:

$$\frac{\sum_{i=1}^n \alpha_i \left(\frac{\beta_i}{\alpha_i} x_i^{2d}\right)}{\sum_{i=1}^n \alpha_i} \ge \left(\prod_{i=1}^n \left(\frac{\beta_i}{\alpha_i} x_i^{2d}\right)^{\alpha_i}\right)^{1/2d}$$

Simplifying the LHS gives $\frac{1}{2d}\sum \beta_i x_i^{2d}$. Rearranging yields:

$$\sum_{i=1}^{n} \beta_i x_i^{2d} \geq 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d} x^{\alpha}$$

By the hypothesis, $\mu \leq 2d \prod (\frac{\beta_i}{\alpha_i})^{\frac{\alpha_i}{2d}}$. Thus, $\sum \beta_i x_i^{2d} \geq \mu x^{\alpha}$.
(⇒) Conversely, assume $f(x)$ is PSD. We prove the condition by contradiction. Suppose:

$$\mu > 2d \left(\prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{1/2d}$$

We construct a test point $x^* \in \mathbb{R}_+^n$ where the AM-GM equality holds. Let $x_i^* = (\frac{\alpha_i}{\beta_i})^{1/2d}$. Evaluating the sum term:

$$\sum_{i=1}^{n} \beta_i (x_i^*)^{2d} = \sum_{i=1}^{n} \beta_i \left(\frac{\alpha_i}{\beta_i} \right) = \sum_{i=1}^{n} \alpha_i = 2d$$

Evaluating the monomial term:

$$(x^*)^{\alpha} = \prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\frac{\alpha_i}{2d}} = \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

Substituting into $f(x^*)$:

$$f(x^*) = 2d - \mu \left(\prod_{i=1}^{n} \left(\frac{\alpha_i}{\beta_i} \right)^{\alpha_i} \right)^{1/2d}$$

Under the assumption that the condition fails (strictly greater μ), we have $f(x^*) < 0$, contradicting that f is PSD. □

We now generalize this result to circuit polynomials, where the tail term is supported by a convex combination of the exponents of the diagonal terms.

Theorem 2. *Let $\alpha(0), \ldots, \alpha(r) \in \mathbb{N}^n$ be exponent vectors. Let $\beta \in \mathbb{N}^n$ be a convex combination of the $\alpha(j)$ vectors, such that $\beta = \sum_{j=0}^{r} \lambda_j \alpha(j)$ with $\lambda_j > 0$ and $\sum \lambda_j = 1$. Consider the polynomial $f(X)$ with coefficients $f_{\alpha(j)} > 0$:*

$$f(X) = \sum_{j=0}^{r} f_{\alpha(j)} X^{\alpha(j)} - f_{\beta} X^{\beta}$$

The polynomial $f(X)$ is PSD for all $X \in \mathbb{R}_+^n$ if and only if:

$$|f_{\beta}| \leq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j}$$

Proof. Let $Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)}$ for $j = 0, \ldots, r$. By the weighted AM-GM inequality:

$$\sum_{j=0}^{r} \lambda_j Y_j \geq \prod_{j=0}^{r} Y_j^{\lambda_j}$$

Substituting Y_j:

$$\sum_{j=0}^{r} \lambda_j \left(\frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} \right) \geq \prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)} \right)^{\lambda_j}$$

The LHS simplifies to $\sum f_{\alpha(j)} X^{\alpha(j)}$. The RHS simplifies to:

$$\left(\prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} \right) X^{\sum \lambda_j \alpha(j)} = \left(\prod_{j=0}^{r} \left(\frac{f_{\alpha(j)}}{\lambda_j} \right)^{\lambda_j} \right) X^{\beta}$$

Thus, if $|f_\beta| \leq \prod (\frac{f_{\alpha(j)}}{\lambda_j})^{\lambda_j}$, it follows that $\sum f_{\alpha(j)} X^{\alpha(j)} \geq |f_\beta| X^\beta \geq f_\beta X^\beta$, proving nonnegativity. The necessary condition follows by selecting X such that the variance of terms Y_j vanishes. □

Definition 1. *A polynomial whose coefficients satisfy the condition in Theorem 2 is called a* ***nonnegative circuit polynomial****. The* ***SONC cone*** $\mathcal{C}_{n,2d}$ *is the conic hull of all nonnegative circuit polynomials of degree* $2d$ *in* n *variables.*

Remark 1. *From the proof of the theorem 2, it is evident that every nonnegative circuit polynomial can be expressed as a mean polynomial of the form* $M_{1,0}(Y, w)$*, where* $Y_j = \frac{f_{\alpha(j)}}{\lambda_j} X^{\alpha(j)}$ *and* $w_j = \lambda_j$*.*

Definition 2. *The* ***SOSONC cone*** $(\Sigma + \mathcal{C})_{n,2d}$ *is the Minkowski sum of the SOS cone* $\Sigma_{n,2d}$ *and the SONC cone* $\mathcal{C}_{n,2d}$*, as introduced in [3].*

3.1 Diagonal-minus-Tail vs Circuit Forms

Theorem 2 implies Theorem 1 as a special case; the converse does not hold in general, so the two results are *not* equivalent. Every D-T form is a circuit polynomial (with a single tail term), but circuit polynomials allow an arbitrary number of diagonal terms whose exponents form a simplex, which is a strictly more general setting.

To see the implication, map the D-T form $f(x) = \sum \beta_i x_i^{2d} - \mu x^\alpha$ to the general framework:

- The diagonal terms correspond to $\alpha(i) = 2de_i$ with coefficients $f_{\alpha(i)} = \beta_i$.
- The tail term exponent is $\beta = \alpha$.
- The convex combination weights are $\lambda_i = \frac{\alpha_i}{2d}$. Hence

$$\sum \lambda_i = \frac{1}{2d} \sum \alpha_i = 1.$$

Applying the condition from Theorem 2:

$$\mu \leq \prod_{i=1}^{n} \left(\frac{\beta_i}{\alpha_i/2d} \right)^{\alpha_i/2d} = \left(\prod_{i=1}^{n} \left(\frac{2d\beta_i}{\alpha_i} \right)^{\alpha_i} \right)^{\frac{1}{2d}}$$

Raising both sides to the power $2d$ yields $\mu^{2d} \leq (2d)^{2d} \prod (\frac{\beta_i}{\alpha_i})^{\alpha_i}$, recovering the condition in Theorem 1.

4 Investigation of $M_{2d,p}(X, \alpha)$ and Connection to SOBS and SOS

In this section, we investigate the containment of the mean polynomial forms $M_{2d,p}(X, \alpha)$ within the cones of Sums of Binomial Squares (SOBS) and Sums of Squares (SOS).

We first address the case where $p = 0$, which corresponds to the geometric mean.

Proposition 2. *The form $M_{2d,0}(X, \alpha)$ is a PSD Diagonal-minus-Tail (D-T) form and is a Sum of Binomial Squares (SOBS) for any $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = 2d$.*

Proof. The fact that $M_{2d,0}$ is a PSD D-T form follows from the definitions established in Section 2. The proof that such forms are SOBS is established in [5]. □

We now consider the cases where $p > 0$. We prove that for small degrees $2d \in \{2, 4, 6\}$, these forms are not only PSD but also SOS.

Theorem 3. *The forms $M_{2d,p}(X, \alpha)$ are Sums of Squares (SOS) for all $n \in \mathbb{N}$ and any $\alpha \in \mathbb{N}^n$ with $|\alpha| = 2d$, for the cases:*

- $d = 1$ *(Quadratic forms), $p \in \{0, 1\}$;*
- $d = 2$ *(Quartic forms), $p \in \{0, 1, 2\}$;*
- $d = 3$ *(Sextic forms), $p \in \{0, 1, 2, 3\}$.*

Proof. We analyze each degree $2d$ separately.
Case $d = 1$ (Degree 2): The forms $M_{2,0}$ and $M_{2,1}$ are PSD quadratic forms. By Hilbert's 1888 Theorem, every nonnegative quadratic form is a sum of squares. Thus, they are SOS.
Case $d = 2$ (Degree 4): For $p = 0$, the form is SOBS (and thus SOS) by Proposition 2. We analyze $p = 1$ and $p = 2$.

- **$p = 1$:** The form is $M_{4,1}(X, \alpha) = \frac{1}{4} \sum \alpha_i X_i^4 - (\frac{1}{4} \sum \alpha_i X_i)^4$.
    1. For $n = 2$ (binary quartics) and $n = 3$ (ternary quartics), Hilbert's Theorem states that all PSD forms are SOS. Since $M_{4,1}$ is PSD, it is SOS.

2. For $n = 4$, we consider the uniform weights $\alpha = (1, 1, 1, 1)$. SDP decomposition using Irene [8] confirms $M_{4,1}(X, (1, 1, 1, 1))_{\text{sos}} = 0$. Moreover, an approximate Gram matrix for this instance is given in [8]:

$$G = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.25 & -0.01 & -0.04 & -0.01 & -0.04 & -0.04 & -0.01 & -0.04 & -0.04 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.05 & -0.01 & 0.01 & 0.01 & -0.04 & 0.01 & 0.01 & -0.02 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.01 & 0.25 & -0.04 & -0.01 & -0.04 & -0.04 & -0.01 & -0.04 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.01 & -0.04 & 0.05 & 0.01 & -0.01 & 0.01 & -0.02 & 0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & 0.01 & -0.01 & 0.01 & 0.05 & -0.01 & -0.02 & 0.01 & 0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.04 & -0.04 & -0.01 & -0.01 & 0.25 & -0.04 & -0.04 & -0.01 & -0.04 \\
0 & 0 & 0 & 0 & 0 & -0.01 & 0.01 & -0.04 & 0.01 & -0.02 & -0.04 & 0.05 & 0.01 & 0.01 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & 0.01 & -0.01 & -0.02 & 0.01 & -0.04 & 0.01 & 0.05 & 0.01 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.02 & -0.04 & 0.01 & 0.01 & -0.01 & 0.01 & 0.01 & 0.05 & -0.01 \\
0 & 0 & 0 & 0 & 0 & -0.04 & -0.04 & -0.04 & -0.04 & -0.04 & -0.04 & -0.01 & -0.01 & -0.01 & 0.25
\end{bmatrix}$$

For non-uniform weights with $|\alpha| = 4$ (e.g., $(2, 1, 1, 0)$), the form effectively involves fewer than 4 variables, reducing to the $n \leq 3$ case, which is SOS.

3. For $n \geq 5$, any partitions of 4 into n integers must contain zeros. The form reduces to a case with $n \leq 4$, which we have shown to be SOS.

- $p = 2$**:** The form is $M_{4,2}(X, \alpha) = \frac{1}{4} \sum \alpha_i X_i^4 - (\frac{1}{4} \sum \alpha_i X_i^2)^2$. Similar to the $p = 1$ case:

1. For $n \leq 3$, it is SOS by Hilbert's Theorem.
2. For $n = 4$ with $\alpha = (1, 1, 1, 1)$, SDP decomposition using Irene [8] confirms $f_{\text{sos}} = 0$. The following is an approximate Gram matrix [8]:

$$G = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.19 & -0 & -0.06 & 0 & -0 & -0.06 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & -0 & 0 & -0 & 0 & -0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & -0 & 0.19 & 0 & -0 & -0.06 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & -0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0 & -0 & -0 & -0 & 0 & 0 & 0 & 0 & 0 & -0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & 0 & -0.06 & -0 & 0 & 0.19 & 0 & 0 & 0 & -0.06 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & -0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -0.06 & 0 & -0.06 & 0 & -0 & -0.06 & 0 & 0 & 0 & 0.19
\end{bmatrix}$$

3. For $n \geq 5$, the support of α limits the effective variables to at most 4, reducing to the previous cases.

Case $d = 3$ **(Degree 6):** We examine $M_{6,p}(X, \alpha)$ for $p \in \{0, \ldots, 3\}$.

1. $n = 2$ **(Binary Sextics):** By Hilbert's Theorem, all PSD binary forms of degree 6 are SOS.

2. $n = 3$ **(Ternary Sextics):**
   - For $p = 0$, it is SOBS (Prop 2).
   - For $p = 1$, we analyze partitions of 6 into 3 parts: $(2, 2, 2)$, $(3, 2, 1)$, $(4, 1, 1)$, $(5, 1, 0)$, $(6, 0, 0)$. The case $\alpha = (2, 2, 2)$ is confirmed to be SOS via SDP decomposition using Irene [8]. Cases with a zero component reduce to $n \leq 2$ (binary sextics, covered by Hilbert's Theorem). The remaining cases $(3, 2, 1)$ and $(4, 1, 1)$ are also confirmed to be SOS via SDP decomposition using Irene [8].
   - For $p = 2$ and $p = 3$, SOS membership for all partitions of 6 into 3 non-zero parts is confirmed via SDP decomposition using Irene [8]; cases with zero components reduce to $n \leq 2$.
3. $n = 4$**:** Relevant partitions of 6 include $(2, 2, 1, 1)$ and $(3, 1, 1, 1)$. SDP decomposition using Irene [8] confirms these specific instances are SOS. Partitions containing zeros reduce to $n \leq 3$.
4. $n \geq 5$**:** Any partition of 6 into $n \geq 6$ parts (e.g., $(1, 1, 1, 1, 1, 1)$) or parts with zeros reduces the effective number of variables.

This completes the proof for $d = 1, 2, 3$. □

Remark 2. *Two open questions remain regarding these forms:*

1. *Does Theorem 3 generalize to any degree d?*
2. *Are these forms contained within the SONC cone for all parameters?*

A clean degree-8 pilot computed with Irene [8] for $d = 4$, $n \in \{3, 4\}$, $p \in \{0, 1, 2, 4\}$, and representative nondegenerate template families yields a consistent pattern: every tested nondegenerate instance admitted an SDP-based SOS decomposition, while SONC feasibility held only for the $p \in \{0, 4\}$ slices. For the same families with $p \in \{1, 2\}$, the SONC signomial constraints remained infeasible under local and global solves and under error bounds 10^{-6}, 10^{-8}, and 10^{-10}. This provides positive evidence for a degree-8 extension of Theorem 3 and negative evidence against a broad SONC containment statement, while still leaving a theorem-level explanation open.

Computational Evidence Snapshot (Degrees 8 and 10)

We summarize the clean Irene-based pilot runs for $d \in \{4, 5\}$, $n \in \{3, 4\}$, $p \in \{0, 1, 2, 4, 5\}$ with template families generated from balanced, mixed, and boundary weight patterns (one-hot degeneracies tracked separately as inconclusive). The clean records are reported in:

- `phase3_runs_clean.jsonl`
- `phase3_runs_clean_d5.jsonl`

Table 1: Pilot classification summary for $d = 4$ (degree 8). Entries report (SDP/SONC/GP) using S = success, T = timeout, F = fail, and I = inconclusive.

template	representative α	support class	n values	$p = 0$	$p = 1$	$p = 2$	$p = d$
uniform	$n = 3$: $(3, 3, 2)$; $n = 4$: $(2, 2, 2, 2)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
mixed	$n = 3$: $(4, 2, 2)$; $n = 4$: $(4, 2, 2, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
boundary	$n = 3$: $(7, 1, 0)$; $n = 4$: $(7, 1, 0, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
sparse	$n = 3$: $(8, 0, 0)$; $n = 4$: $(8, 0, 0, 0)$	degenerate	$\{3, 4\}$	I/I/I	I/I/I	I/I/I	I/I/I
$d \geq 6$	–	placeholder	$\{3, 4\}$ planned	deferred	deferred	deferred	deferred

Table 2: Pilot classification summary for $d = 5$ (degree 10). Entries report (SDP/SONC/GP) using S = success, T = timeout, F = fail, and I = inconclusive.

template	representative α	support class	n values	$p = 0$	$p = 1$	$p = 2$	$p = d$
uniform	$n = 3$: $(4, 3, 3)$; $n = 4$: $(3, 3, 2, 2)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
mixed	$n = 3$: $(5, 3, 2)$; $n = 4$: $(5, 3, 2, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
boundary	$n = 3$: $(9, 1, 0)$; $n = 4$: $(9, 1, 0, 0)$	simplex-like	$\{3, 4\}$	S/S/S	S/F/S	S/F/S	S/S/S
sparse	$n = 3$: $(10, 0, 0)$; $n = 4$: $(10, 0, 0, 0)$	degenerate	$\{3, 4\}$	I/I/I	I/I/I	I/I/I	I/I/I
$d \geq 6$	–	placeholder	$\{3, 4\}$ planned	deferred	deferred	deferred	deferred

- `phase3_classification_table.csv`

For both $d = 4$ and $d = 5$, every nondegenerate tested instance admitted an SDP-based SOS decomposition in our current grid. In contrast, SONC feasibility is support-sensitive: the nondegenerate families with $p \in \{1, 2\}$ that fail at degree 8 remain infeasible under local/global SONC solves and tighter tolerance settings, while GP relaxations remain feasible on all tested nondegenerate families in the same grid. This pattern supports a refined view in which SOS positivity extends robustly on the tested families, whereas SONC membership requires stronger structural conditions than the broad conjectural form. Tables 1 and 2 summarize the refreshed classification output using the corrected support labels. We retain a placeholder row for $d \geq 6$ in the manuscript tables; a completed $d = 6$ L-C2 SONC/GP slice and a staged $d = 6$ L-C1 SOS/SDP extension are tracked in companion artifacts. For $d = 6$, $n = 3$ uniform L-C1 slices are successful in the tested grid, whereas $n = 4$ L-C1 slices exhibit timeout-dominated behavior across uniform, boundary, and mixed templates (including $p = 6$ cross-template probes) under budgets up to 1800s. We therefore mark $d = 6$ L-C1 as computationally deferred in the current manuscript tables. Additional CX-1 tractability probes indicate a solver-scaling ceiling for the current CVXOPT SDP pipeline at $n \leq 4$ for L-C1 families: d=4 probes at $n = 5, 6$ (boundary/mixed templates) timeout at 120s, d=5 at $n = 5$ (boundary probe) times out at 900s, and even d=3 at $n = 5$ requires 272s for a single successful boundary probe. This supports treating $n \geq 5$ high-degree sparse-family extensions as computationally deferred pending specialized solver paths.

Deferred higher-degree slice. The $d \geq 6$ rows are retained as placeholders for a future extension of the same protocol. No higher-degree computations are included in the present manuscript snapshot.

Claim-Status Synthesis (Frozen $d = 4, d = 5$ Slice)

Using the frozen clean artifacts and refreshed classification outputs (listed in the reproducibility note below), we classify the current computational evidence as follows.

- **Confirmed (within tested grid).** For nondegenerate tested families at $d = 4, 5$ with $n \in \{3, 4\}$, SDP/SOS and GP outcomes are consistently feasible in the current runs.
- **Weakened (scope-limited).** Any broad SONC containment formulation for the same families is not supported uniformly; the evidence favors a support-sensitive SONC statement rather than a degree-only claim.
- **Negative (robust failures).** For nondegenerate families with $p \in \{1, 2\}$, SONC infeasibility persists under local/global solves and tighter tolerances in both degree slices, while SDP/SOS and GP remain feasible.

This synthesis is tied to the classification tables in Tables 1 and 2, with a partial higher-degree extension from the completed $d = 6$ L-C2 SONC/GP slice and a deferred-status $d = 6$ L-C1 SOS/SDP record.

Reproducibility note. The tables above are derived from the frozen $d = 4, d = 5$ artifacts:

- `phase3_runs_clean.jsonl`
- `phase3_runs_clean_d5.jsonl`
- `phase3_classification_table.csv`
- `phase3_classification_table.md`

A partial higher-degree extension is available for $d = 6$. The completed L-C2 SONC/GP slice and staged L-C1 SOS/SDP artifacts are:

- `phase3_runs_clean_d6_lc2.jsonl`
- `phase3_pilot_summary_clean_d6_lc2.md`
- `phase3_sonc_diagnostics_d6_lc2.jsonl`
- `phase3_runs_clean_d6_lc1_batch1.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch1.md`
- `phase3_runs_clean_d6_lc1_batch2.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch2.md`
- `phase3_runs_clean_d6_lc1_batch3.jsonl`

- `phase3_pilot_summary_clean_d6_lc1_batch3.md`
- `phase3_runs_clean_d6_lc1_batch4_boundary.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch4_boundary.md`
- `phase3_runs_clean_d6_lc1_batch5_mixed.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch5_mixed.md`
- `phase3_runs_clean_d6_lc1_batch6_uniform_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch6_uniform_p6.md`
- `phase3_runs_clean_d6_lc1_batch7_boundary_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch7_boundary_p6.md`
- `phase3_runs_clean_d6_lc1_batch8_mixed_p6.jsonl`
- `phase3_pilot_summary_clean_d6_lc1_batch8_mixed_p6.md`
- `phase3_runs_clean_cx1_d4_n5.jsonl`
- `phase3_runs_clean_cx1_d4_n6.jsonl`
- `phase3_runs_clean_cx1_batch1_d5_n5_boundary_mixed.jsonl`
- `phase3_runs_clean_cx1_d3_n5_probe.jsonl`
- `phase3_pilot_summary_cx1_d3_n5_probe.md`

The manuscript snippet `results/phase3_classification_tables.tex` is generated from the refreshed CSV by the table-builder script in `scripts/`. This keeps updates to the frozen $d = 4, d = 5$ slice and staged $d \geq 6$ extensions reproducible without manual table editing.

Follow-up evidence. The CX-1 tractability probe is completed and currently supports a solver-scaling bottleneck interpretation (rather than a new non-SOS counterexample) for sparse-family $n \geq 5$ slices under the present CVXOPT SDP setup. Cross-degree transform stress tests (CX-2) and specialized-solver escalation paths for deferred $d = 6/n \geq 5$ SOS/SDP slices remain in progress.

5 Beyond the SOSONC Cone

In this section, we move towards a comprehensive understanding of the general form $M_{q,p}(Y, w)$ beyond the established SOSONC cone. We examine specific classical forms—the Choi-Lam form Q and the Robinson forms R and $\hat{R}$—and establish their representation as mean polynomials.

5.1 Specific Examples

We consider the following forms in $\mathbb{R}[X, Y, Z, W]$:

- The Choi-Lam form Q:

$$Q(X, Y, Z, W) = X^4 + Y^4 + Z^4 + W^4 - 4XYZW$$

- The Robinson form R:

$$\begin{aligned} R(X, Y, Z) = X^6 + Y^6 + Z^6 \\ - (X^4Y^2 + X^4Z^2 + X^2Y^4 + X^2Z^4 + Y^4Z^2 + Y^2Z^4) \\ + 3X^2Y^2Z^2 \end{aligned}$$

- The second Robinson form $\hat{R}$:

$$\begin{aligned} \hat{R}(X, Y, Z, W) = X^2(X - W)^2 + Y^2(Y - W)^2 + Z^2(Z - W)^2 \\ + 2XYZ(X + Y + Z - 2W) \end{aligned}$$

It is observed in [12] that Q is a Sum of Nonnegative Circuit (SONC) polynomial. However, using the Newton polytope method, one can prove that $Q \in \mathcal{P}_{4,4} \setminus \Sigma_{4,4}$ (it is PSD but not SOS).

Remarkably, there exists a linear relationship between Q and $\hat{R}$ as noted by Reznick [11]:

$$\hat{R}(X - W, Y - W, Z - W, X + Y + Z - W) = 2Q(X, Y, Z, W) \tag{1}$$

Since the PSD and SOS cones are closed under linear transformations, the identity above shows that $\hat{R}$ is PSD (respectively SOS) if and only if Q is PSD (respectively SOS). Consequently, $\hat{R}$ also serves as an example of a form in $\mathcal{P}_{4,4} \setminus \Sigma_{4,4}$.

5.2 Mean Polynomial Representations

We now establish the representation of these forms as instances of mean polynomials $M_{q,p}$.

Proposition 3. *The Choi-Lam form Q and the Robinson form $\hat{R}$ can be expressed as mean polynomials of order* 1 *and* 0 *(arithmetic minus geometric means).*

Proof. For the Choi-Lam form Q, we observe:

$$Q(X, Y, Z, W) = 4\left(\frac{X^4 + Y^4 + Z^4 + W^4}{4} - \sqrt[4]{X^4Y^4Z^4W^4}\right)$$

This corresponds exactly to the mean form definition with variables being the fourth powers and uniform weights:

$$Q(X, Y, Z, W) = 4M_{1,0}((X^4, Y^4, Z^4, W^4), (1, 1, 1, 1))$$

For the Robinson form $\hat{R}$, applying the inverse of the linear transformation in (1), we derive:

$$\hat{R}(X,Y,Z,W) = 8M_{1,0}((\hat{X}^4, \hat{Y}^4, \hat{Z}^4, \hat{W}^4), (1,1,1,1))$$

where

$$(\hat{X}, \hat{Y}, \hat{Z}, \hat{W}) = \frac{1}{2}(X + W - Y - Z,\ Y + W - X - Z,\\ Z + W - X - Y,\ W - (X + Y + Z)).$$

□

This representation is significant for distinguishing the cone of mean polynomials from the SOSONC cone. While Q is SONC (and thus in the SOSONC cone) because it is a mean of monomials in the standard basis, $\hat{R}$ is a mean of *linear forms*. Since the SONC property is not invariant under general linear coordinate changes, $\hat{R}$ serves as a separating form: it belongs to the cone generated by generalized mean polynomials (means of linear forms) but lies outside the standard SOSONC cone (see [12, page 31 and Lemma 4.2.5, page 107]).

6 Positivestellensatz and relation to the cone of $2d$ powers

This section is reserved for a follow-up paper.

7 Application to polynomial optimization

This section is reserved for a follow-up paper.

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